
Slide 1: Title Page

"Hello professor Today, I will present a structure review of the journal paper titled:
'Convolutional autoencoder-based SOH estimation of lithium-ion batteries using electrochemical impedance spectroscopy.'"

Slide 2: Outline

"Here is the outline of my presentation. I will start with the introduction and problems with current methods. Then, I will explain the proposed methodology, specifically the **CAE-DNN model**. After that, I will show the experimental results and finish with a critical review and future work."

Slide 3: Introduction

"Let's start with why this research is important. Lithium-ion batteries are everywhere, especially in EVs. We need to monitor their **State of Health (SOH)** accurately for safety. However, traditional methods have problems.

Please look at Figure 1(a) on the left. This is an Equivalent Circuit Model. It requires complex physical parameters and is hard to identify.

On the right, Figure 1(b), shows a Nyquist plot. Traditional methods use manual feature extraction on this data, which is slow and can have human errors. So, we need an automatic way to do this."

Slide 4: Why EIS Data?

"Why do we use **EIS data**? EIS stands for Electrochemical Impedance Spectroscopy. It allows us to see 'inside' the battery.

Please look at Figure 2. These are Nyquist plots for batteries at different temperatures: 25, 35, and 45 degrees Celsius.

Pay attention to the curves. You can see the blue circles (90% SOH) and yellow triangles (80% SOH) shift to the right compared to the green circles (100% SOH). This shows that as battery health drops, the impedance changes. Because this pattern is clear, we can use a

Deep Learning model called a Convolutional Autoencoder (CAE) to learn these features automatically."

Slide 5: Objectives & Overview

"The main objective of this paper is to build a model called **CAE-DNN**. This model estimates SOH using only EIS data without manual calculations.

Please look at the workflow in Figure 3.

First, on the left, we gather EIS data from the batteries.

Second, in the middle, we preprocess this data.

Finally, on the right, we feed it into the CAE-DNN model to get the SOH estimation. The key innovation here is transforming raw data into 2D images, which I will explain next."

Slide 6: Data Preprocessing

"This is the most important part: Data Preprocessing.

Raw EIS data is just a list of numbers. We measure impedance at 60 different frequencies.

Please look at Figure 4.

We take these raw numbers and normalize them. Then, we add statistical values like mean and standard deviation.

Finally, we reshape this data into an 8x8 matrix with 2 channels—one for Real impedance and one for Imaginary impedance. This turns the data into a '2D Image'. This allows us to use Convolutional layers, which are very good at processing images."

Slide 7: Model Architecture (CAE-DNN)

"Now, let's look at the model architecture in **Figure 6** (labeled here as the detailed diagram). It has two stages.

Stage 1 is on the left, the **Convolutional Autoencoder**. It takes the 8x8 image as input. The Encoder compresses the image into a small latent vector of size 256. This process extracts the most important features and removes noise automatically.

Stage 2 is on the right, the **Deep Neural Network (DNN)**. It takes that latent vector and passes it through 4 dense layers. The final output is the estimated charge capacity, which gives us the SOH."

Slide 8: Experimental Setup

"For the experiment, they used Eunicell LR2032 coin cells.

They tested the batteries at three temperatures: 25, 35, and 45 degrees Celsius.

A key point here is the data splitting. They used 6 cells for training, mostly at 25 degrees. But they tested on 2 cells at 35 and 45 degrees. This is to check if the model works well in hot environments.

Table 3 lists the parameters. We compared our CAE-DNN model against 5 baseline methods, including GPR, Random Forest, and a standard CNN."

Slide 9: Quantitative Results

"Here are the numerical results. Please look at Table 4.

This table compares the errors (RMSE) and accuracy (R^2) of all models.

The bottom row shows the CAE-DNN. It performed the best.

At 35 degrees, it achieved an R^2 of 0.9657.

At 45 degrees, the result is very significant. The standard CNN (second from bottom) failed, with an R^2 of only 0.35. However, the CAE-DNN remained accurate with an R^2 of 0.97. This proves the model is robust at high temperatures."

Slide 10: Visual Results

"Let's visualize these results. Please look at the graphs in Figure 8.

These show the estimated SOH (colored lines) versus the measured SOH (black line) at 45 degrees Celsius.

Look at the bottom-right graph. The red line is the CAE-DNN model. It follows the black line very closely from start to finish.

Now, compare it to the bottom-middle graph. The blue line is the standard CNN. It drops way too fast and does not match the real health. This visually confirms that CAE-DNN is more accurate than other methods."

Slide 11: Discussion and Ablation Study

"The authors also did an ablation study to find the best settings, as shown in Table 5.

They tested different numbers of filters and dropout rates. They found that using 4 filters and a dropout rate of 0.03 gave the lowest error.

Why is CAE better than a standard CNN?

The Autoencoder uses unsupervised learning. This helps it learn the structure of the data better and removes noise from the EIS measurements, making it more stable than a standard CNN."

Slide 12: Conclusion

"To conclude, this paper successfully proposed the **CAE-DNN model** for battery health estimation.

The key takeaway is that transforming EIS data into 2D images allows the model to learn hidden features automatically.

The results show it outperforms traditional methods like GPR and CNN, especially at high temperatures like 45 degrees Celsius."

Slide 13: Critical Review & Future Work

"Finally, here is my critical review.

Strengths: The idea of converting data into images is very clever. It makes the model robust against temperature changes.

Limitations: The paper only tested batteries that were **fully charged (100% SOC)**. In real-world EVs, we cannot always wait for a full charge to check battery health.

Future Work: Next, we should test this model using data from **partial charging**, for example, at 50% battery level. We could also try newer AI models like **Transformers** to see if they improve accuracy further."

Slide 14: Thank You

"That concludes my presentation. Thank you for your attention."